

Unit 4

Multidimensional Analysis and Descriptive Mining of Complex Data Objects –
Spatial Data Mining – Multimedia Data Mining – Text Mining – Mining the
World Wide Web.

What is Spatial Data Mining?

Spatial data mining is a specialized subfield of data mining that deals with extracting knowledge from spatial data.

Spatial data refers to data that is associated with a particular location or geography.

Examples of spatial data include maps, satellite images, GPS data, and other geospatial information.

Spatial data mining involves analyzing and discovering patterns, relationships, and trends in this data to gain insights and make informed decisions.

The use of spatial data mining has become increasingly important in various fields, such as logistics, environmental science, urban planning, transportation, and public health.

By analyzing spatial data, researchers and data mining professionals can identify correlations, predict future events, and make informed decisions that can have a significant impact.

For instance, a transportation company can optimize its delivery routes for faster and more efficient deliveries using spatial data mining techniques.

They can analyze their delivery data along with other spatial data, such as traffic flow, road network, and weather patterns, to identify the most efficient routes for each delivery.

Spatial data mining tasks

These are the primary tasks of spatial data mining.

Classification:

Classification determines a set of rules which find the class of the specified object as per its attributes.

Association rules:

Association rules determine rules from the data sets, and it describes patterns that are usually in the database.

Characteristic rules:

Characteristic rules describe some parts of the data set.

Discriminate rules:

As the name suggests, discriminate rules describe the differences between two parts of the database, such as calculating the difference between two cities as per employment rate.

Types of Spatial Data

Different types of spatial data are used in spatial data mining. These include point data, line data, and polygon data.

- **Point Data**

- Point data represents a single location or a set of locations on a map. Each point is defined by its x and y coordinates, representing its position in the geographic space. Point data is commonly used to represent geographic features such as cities, landmarks, or specific locations of interest. Examples of point data in transportation include delivery locations, bus stops, or railway stations.

- **Line Data**

- Line data represents a linear feature, such as a road, a river, or a pipeline, on a map. Each line is defined by a set of vertices, which represent the start and end points of the line. Line data is commonly used to represent transportation networks, such as roads, highways, or railways. Line data is also used in other areas, such as hydrology, geology, or ecology, to represent streams, faults, or animal migration routes.
- **Polygon Data**
 - Polygon data represents a closed shape or an area on a map. Each polygon is defined by a set of vertices that connect to form a closed boundary. Polygon data is commonly used to represent administrative boundaries, land use, or demographic data. In transportation, polygon data can be used to represent areas of interest, such as delivery zones or traffic zones.

Advantages of Spatial Data Mining

- **Insight Into Geographical Patterns:** Spatial statistics assists in identifying such features that would otherwise, lay undetected, by enabling organizations and researchers, to identify trends concerning the area.
- **Better Decision Making:** Spatial data mining is thus applicable in areas such as urban planning, environmental management, and logistics resources in organizations to make a wise decision.
- **Enhanced Visualization:** The data collected at the different spatial levels can be presented and represented in maps, which gives a better view of the trends and patterns.

Disadvantages of Spatial Data Mining

- **Complexity of Data:** This means that the raw data may be huge and intricate, making the application of complex algorithms necessary together with huge computing facilities for data processing.
- **Data Inaccuracy:** Some mistakes can be made while collecting the data like wrong geographical coordinates of any place and wrong conclusions can be drawn as a result of that.

Temporal Mining

Temporal data refers to the extraction of simple, non-trivial and potentially useful abstract information from large collection of [temporal data](#). It is concerned with the analysis of temporal data and for finding temporal patterns and regularities in sets of temporal data tasks of temporal data mining are –

- Data Characterization and Comparison
- Cluster Analysis
- Classification
- Association rules
- Prediction and Trend Analysis
- Pattern Analysis

Advantages of Temporal Data Mining

- **Trend Identification:** Forecasting is also beneficial in identification of temporal factors such as seasonality, cycle and trends, or generation of long-term shifts.
- **Forecasting:** It can be used in a predictive way to help the organization on how they can be prepared to face the future outcomes or trends.
- **Anomaly Detection:** One of the strengths of temporal data mining is that it is able to identify patterns that change frequently especially where there

is evidence of sharp changes which may point to either an event or a problem.

Disadvantages of Temporal Data Mining

- **Handling Data Complexity:** A lot of times, time-series data will be some kind of dependent relationship with time points, and this makes analysis very difficult.
- **Requires Large Historical Data:** The problem of temporal data mining is that it might take a large amount of historical data to find useful patterns which is not always feasible.

Difference Between Spatial and Temporal Data Mining

Spatial data mining	Temporal data mining
It requires space.	It requires time.
Spatial mining is the extraction of knowledge/spatial relationship and interesting measures that are not explicitly stored in spatial database.	Temporal mining is the extraction of knowledge about occurrence of an event whether they follow Cyclic , Random ,Seasonal variations etc.
It deals with spatial (location , Geo-referenced) data.	It deals with implicit or explicit Temporal content , from large quantities of data.
Spatial databases reverses spatial objects derived by spatial data. types and spatial association among such objects.	Temporal data mining comprises the subject as well as its utilization in modification of fields.

Spatial data mining	Temporal data mining
It includes finding characteristic rules, discriminant rules, association rules and evaluation rules etc. It is the method of identifying unusual and unexplored data but useful models from spatial databases.	It aims at mining new and unknown knowledge, which takes into account the temporal aspects of data. It deals with useful knowledge from temporal data.
Examples – Determining hotspots, Unusual locations.	Examples – An association rule which looks like – “Any Person who buys a car also buys steering lock”.

What is Multimedia Data Mining?

Multimedia mining is a subfield of data mining that is used to find interesting information of implicit knowledge from multimedia databases. Mining in multimedia is referred to as automatic annotation or annotation mining. Mining multimedia data requires two or more data types, such as text and video or text video and audio.

Multimedia data mining is an interdisciplinary field that integrates image processing and understanding, computer vision, data mining, and pattern recognition. Multimedia data mining discovers interesting patterns from multimedia databases that store and manage large collections of multimedia objects, including image data, video data, audio data, sequence data and hypertext data containing text, text markups, and linkages. Issues in multimedia data mining include *content-based retrieval and similarity search, generalization and*

multidimensional analysis. Multimedia data cubes contain additional dimensions and measures for multimedia information.

The framework that manages different types of multimedia data stored, delivered, and utilized in different ways is known as a multimedia database management system. There are three classes of multimedia databases: static, dynamic, and dimensional media. The content of the Multimedia Database management system is as follows:

- **Media data:** The actual data representing an object.
- **Media format data:** Information such as sampling rate, resolution, encoding scheme etc., about the format of the media data after it goes through the acquisition, processing and encoding phase.
- **Media keyword data:** Keywords description relating to the generation of data. It is also known as content descriptive data. Example: date, time and place of recording.
- **Media feature data:** Content dependent data such as the distribution of colours, kinds of texture and different shapes present in data.

Types of Multimedia Applications

Types of multimedia applications based on data management characteristics are:

1. **Repository applications:** A Large amount of multimedia data and meta-data (Media format data, Media keyword data, Media feature data) that is stored for retrieval purposes, e.g., Repository of satellite images, engineering drawings, radiology scanned pictures.
2. **Presentation applications:** They involve delivering multimedia data subject to temporal constraints. Optimal viewing or listening requires DBMS to deliver data at a certain rate, offering the quality of service above a certain threshold. Here data is processed as it is delivered. Example: Annotating of video and audio data, real-time editing analysis.

3. **Collaborative work using multimedia information** involves executing a complex task by merging drawings and changing notifications. Example: Intelligent healthcare network.

Challenges with Multimedia Database

There are many challenges to multimedia databases, such as:

1. **Modelling:** Working in this area can improve database versus information retrieval techniques; thus, documents constitute a specialized area and deserve special consideration.
2. **Design:** The conceptual, logical and physical design of multimedia databases has not yet been addressed fully as performance and tuning issues at each level are far more complex as they consist of a variety of formats like JPEG, GIF, PNG, MPEG, which is not easy to convert from one form to another.
3. **Storage:** Storage of multimedia database on any standard disk presents the problem of representation, compression, mapping to device hierarchies, archiving and buffering during input-output operation. In DBMS, a BLOB (Binary Large Object) facility allows untyped bitmaps to be stored and retrieved.
4. **Performance:** Physical limitations dominate an application involving video playback or audio-video synchronization. The use of parallel processing may alleviate some problems, but such techniques are not yet fully developed. Apart from this, a multimedia database consumes a lot of processing time and bandwidth.
5. **Queries and retrieval:** For multimedia data like images, video, and audio accessing data through query open up many issues like efficient query formulation, query execution and optimization, which need to be worked upon.

Where is Multimedia Database Applied?

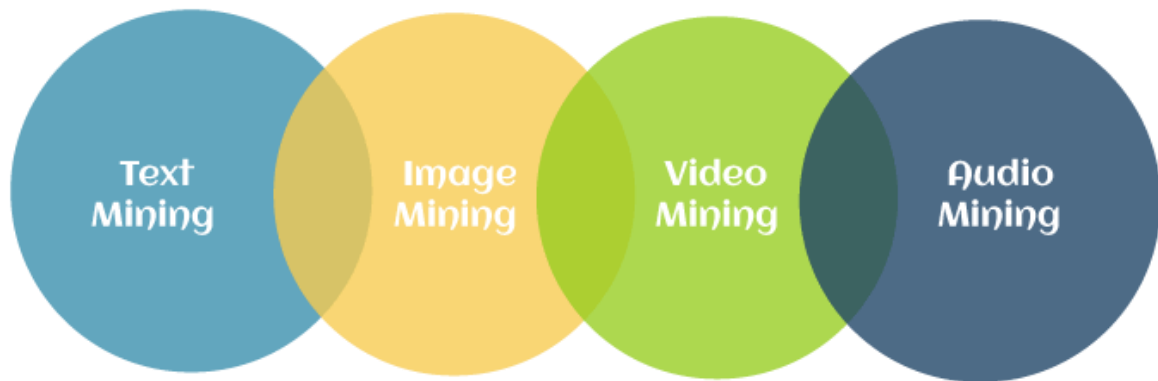
Below are the following areas where a multimedia database is applied, such as:

- **Documents and record management:** Industries and businesses keep detailed records and various documents. For example, insurance claim records.
- **Knowledge dissemination:** Multimedia database is a very effective tool for knowledge dissemination in terms of providing several resources. For example, electronic books.
- **Education and training:** Computer-aided learning materials can be designed using multimedia sources which are nowadays very popular sources of learning. Example: Digital libraries.
- **Travelling:** Marketing, advertising, retailing, entertainment and travel. For example, a virtual tour of cities.
- **Real-time control and monitoring:** With active database technology, multimedia presentation of information can effectively monitor and control complex tasks. For example, manufacturing operation control.

Categories of Multimedia Data Mining

Multimedia data mining is classified into two broad categories: static and dynamic media. *Static media* contains text (digital library, creating SMS and MMS) and images (photos and medical images). *Dynamic media* contains Audio (music and MP3 sounds) and Video (movies). The below image shows the categories of multimedia data mining.

Categories of Multimedia Data Mining



1. Text Mining

Text is the foremost general medium for the proper exchange of information. Text Mining evaluates a huge amount of usual language text and detects exact patterns to find useful information. Text Mining also referred to as text data mining, is used to find meaningful information from unstructured texts from various sources.

2. Image Mining

Image mining systems can discover meaningful information or image patterns from a huge collection of images. Image mining determines how low-level pixel representation consists of a raw image or image sequence that can be handled to recognize high-level spatial objects and relationships. It includes digital image processing, image understanding, database, AI, etc.

3. Video Mining

Video mining is unsubstantiated to find interesting patterns from many video data; multimedia data is video data such as text, image, metadata, visuals and audio. It is commonly used in security and surveillance, entertainment, medicine, sports and education programs. The processing is indexing, automatic segmentation, content-based retrieval, classification and detecting triggers.

4. Audio Mining

Audio mining plays an important role in multimedia applications, is a technique by which the content of an audio signal can be automatically searched, analyzed and rotten with wavelet transformation. It is generally used in automatic speech recognition, where the analysis efforts to find any speech within the audio. Band energy, frequency centroid, zero-crossing rate, pitch period and bandwidth are often used for audio processing.

Application of Multimedia Mining

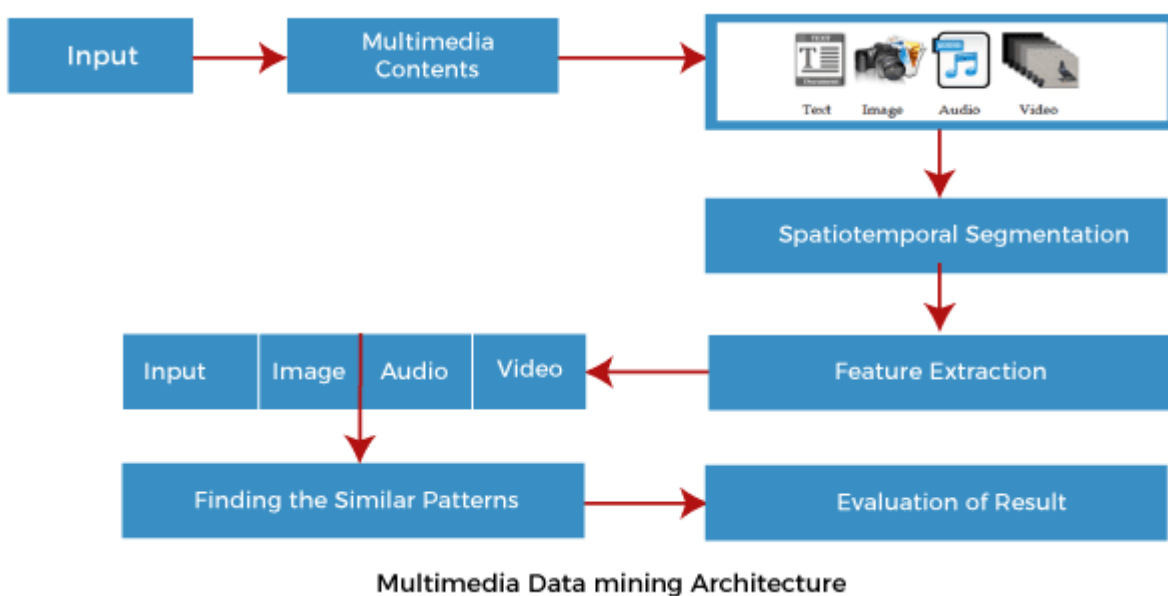
There are different kinds of applications of multimedia data mining, some of which are as follows:

- **Digital Library:** The collection of digital data is stored and maintained in a digital library, which is essential to convert different digital data formats into text, images, video, audio, etc.
- **Traffic Video Sequences:** To determine important but previously unidentified knowledge from the traffic video sequences, detailed analysis and mining are to be performed based on vehicle identification, traffic flow, and queue temporal relations of the vehicle at an intersection. This provides an economic approach for regular traffic monitoring processes.
- **Medical Analysis:** Multimedia mining is primarily used in the medical field, particularly for analyzing medical images. Various data mining techniques are used for image classification. Examples, Automatic 3D delineation of highly aggressive brain tumours, Automatic localization and identification of vertebrae in 3D CT scans, MRI Scans, ECG and X-Ray.
- **Customer Perception:** It contains details about customers' opinions, products or services, customers complaints, customers preferences, and the level of customer satisfaction with products or services, which are collected together. The audio data serve as topic detection, resource assignment and evaluation of the quality of services. Many companies have call centres that receive telephone calls from customers.

- **Media Making and Broadcasting:** Radio stations and TV channels create broadcasting companies, and multimedia mining can be applied to monitor their content to search for more efficient approaches and improve their quality.
- **Surveillance system:** It consists of collecting, analyzing, summarizing audio, video or audiovisual information about specific areas like government organizations, multi-national companies, shopping malls, banks, forests, agricultural areas and, highways etc. The main use of this technology in the field of security; hence it can be utilized by military, police and private companies since they provide security services.

Architecture for Multimedia Data Mining

Multimedia mining architecture is given in the below image. The architecture has several components. Important components are Input, Multimedia Content, Spatiotemporal Segmentation, Feature Extraction, Finding similar Patterns, and Evaluation of Results.



1. **The input** stage comprises a multimedia database used to find the patterns and perform the data mining.

2. ***Multimedia Content*** is the data selection stage that requires the user to select the databases, subset of fields, or data for data mining.
3. ***Spatio-temporal segmentation*** is nothing but moving objects in image sequences in the videos, and it is useful for object segmentation.
4. ***Feature extraction*** is the preprocessing step that involves integrating data from various sources and making choices regarding characterizing or coding certain data fields to serve when inputs to the pattern-finding stage. Such representation of choices is required because certain fields could include data at various levels and are not considered for finding a similar pattern stage. In MDM, the preprocessing stage is significant since the unstructured nature of multimedia records.
5. ***Finding a similar pattern*** stage is the heart of the whole data mining process. The hidden patterns and trends in the data are basically uncovered in this stage. Some approaches to finding similar pattern stages contain association, classification, clustering, regression, time-series analysis and visualization.
6. ***Evaluation of Results*** is a data mining process used to evaluate the results, and this is important to determine whether the prior stage must be revisited or not. This stage consists of reporting and using the extracted knowledge to produce new actions, products, services, or marketing strategies.

What is Text Mining?

Text mining is a component of data mining that deals specifically with unstructured text data. It involves the use of natural language processing (NLP) techniques to extract useful information and insights from large amounts of unstructured text data. Text mining can be used as a preprocessing step for data mining or as a standalone process for specific tasks.

Why is Text Mining Important?

Text mining is widely used in various fields, such as [natural language processing](#), information retrieval, and social media analysis. It has become an essential tool for organizations to extract insights from unstructured text data and make data-driven decisions.

Common Methods for Analyzing Text Mining

- **Text Summarization:** To extract its partial content and reflect its whole content automatically.
- **Text Categorization:** To assign a category to the text among categories predefined by users.
- **Text Clustering:** To segment texts into several clusters, depending on the substantial relevance.

Text Mining Techniques

Information Retrieval

In the process of Information retrieval, we try to process the available documents and the text data into a structured form so, that we can apply different pattern recognition and analytical processes. It is a process of extracting relevant and associated patterns according to a given set of words or text documents.

Information Extraction

It is a process of extracting meaningful words from documents.

- [**Feature Extraction**](#) – In this process, we try to develop some new features from existing ones. This objective can be achieved by parsing an existing feature or combining two or more features based on some mathematical operation.
- [**Feature Selection**](#) – In this process, we try to reduce the dimensionality of the dataset which is generally a common issue while dealing with the text data by selecting a subset of features from the whole dataset.

Natural Language Processing

Natural Language Processing includes tasks that are accomplished by using Machine Learning and Deep Learning methodologies. It concerns the automatic processing and analysis of unstructured text information.

Named Entity Recognition (NER): Identifying and classifying named entities such as people, organizations, and locations in text data.

Sentiment Analysis: Identifying and extracting the sentiment (e.g. positive, negative, neutral) of text data.

Text Summarization: Creating a condensed version of a text document that captures the main points.

Advantages of Text Mining

- **Large Amounts of Data:** Text mining allows organizations to extract insights from large amounts of unstructured text data.
- **Variety of Applications:** Text mining has a wide range of applications, including sentiment analysis, named entity recognition, and topic modeling.
- **Improved Decision Making**
- **Cost-effective:** Text mining can be a cost-effective way, as it eliminates the need for manual data entry.

Disadvantages of Text Mining

- **Complexity:** Text mining can be a complex process requiring advanced skills in natural language processing and machine learning.
- **Quality of Data:** The quality of text data can vary, affecting the accuracy of the insights extracted from text mining.

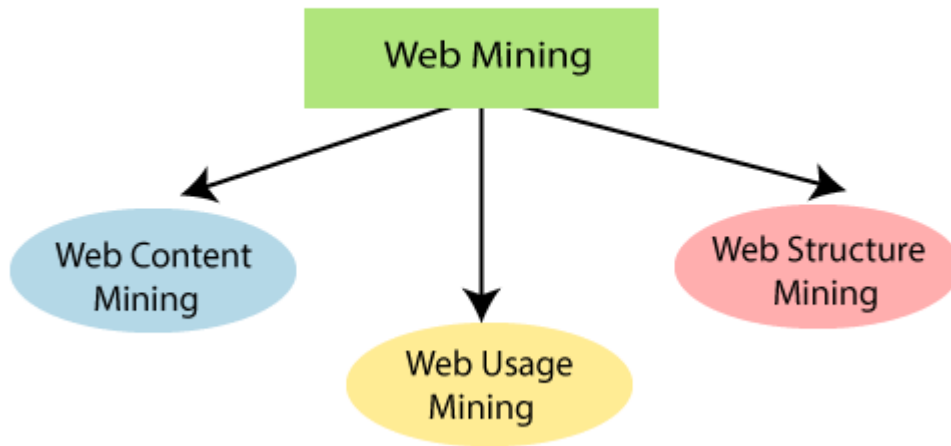
- **High Computational Cost:** Text mining requires high computational resources, and it may be difficult for smaller organizations to afford the technology.
- **Limited to Text Data:** Text mining is limited to extracting insights from unstructured text data and cannot be used with other data types.
- **Noise in text mining results:** Text mining of documents may result in mistakes. It's possible to find false links or to miss others. In most situations, if the noise (error rate) is sufficiently low, the benefits of automation exceed the chance of a larger mistake than that produced by a human reader.
- **Lack of transparency:** Text mining is frequently viewed as a mysterious process where large corpora of text documents are input and new information is produced. Text mining is in fact opaque when researchers lack the technical know-how or expertise to comprehend how it operates, or when they lack access to corpora or text mining tools.

What is Web Mining?

Web mining can widely be seen as the application of adapted data mining techniques to the web, whereas data mining is defined as the application of the algorithm to discover patterns on mostly structured data embedded into a knowledge discovery process. Web mining has a distinctive property to provide a set of various data types. The web has multiple aspects that yield different approaches for the mining process, such as web pages consist of text, web pages are linked via hyperlinks, and user activity can be monitored via web server logs. These three features lead to the differentiation between the three areas are web content mining, web structure mining, web usage mining.

There are three types of data mining:

Types of Web Mining



1. Web Content Mining:

Web content mining can be used to extract useful data, information, knowledge from the web page content. In web content mining, each web page is considered as an individual document. The individual can take advantage of the semi-structured nature of web pages, as HTML provides information that concerns not only the layout but also logical structure. The primary task of content mining is data extraction, where structured data is extracted from unstructured websites. The objective is to facilitate data aggregation over various web sites by using the extracted structured data. Web content mining can be utilized to distinguish topics on the web. For Example, if any user searches for a specific task on the search engine, then the user will get a list of suggestions.

2. Web Structured Mining:

The web structure mining can be used to find the link structure of hyperlink. It is used to identify that data either link the web pages or direct link network. In Web Structure Mining, an individual considers the web as a directed graph, with the web pages being the vertices that are associated with hyperlinks. The most important application in this regard is the Google search engine, which estimates the ranking of its outcomes primarily with the PageRank algorithm. It

characterizes a page to be exceptionally relevant when frequently connected by other highly related pages. Structure and content mining methodologies are usually combined. For example, web structured mining can be beneficial to organizations to regulate the network between two commercial sites.

3. Web Usage Mining:

Web usage mining is used to extract useful data, information, knowledge from the weblog records, and assists in recognizing the user access patterns for web pages. In Mining, the usage of web resources, the individual is thinking about records of requests of visitors of a website, that are often collected as web server logs. While the content and structure of the collection of web pages follow the intentions of the authors of the pages, the individual requests demonstrate how the consumers see these pages. Web usage mining may disclose relationships that were not proposed by the creator of the pages.

Application of Web Mining:

Web mining has an extensive application because of various uses of the web. The list of some applications of web mining is given below.

- Marketing and conversion tool
- Data analysis on website and application accomplishment.
- Audience behavior analysis
- Advertising and campaign accomplishment analysis.
- Testing and analysis of a site.

SECTION A

1. What is text mining?
2. Define temporal mining.
3. Define spatial mining.
4. Name the categories of multimedia data mining.

5. Define web mining.
6. What is web content mining.
7. Define web usage mining.
8. Define web structured mining.
9. What is information retrieval.
10. Define information extraction.
11. Define Natural Language Processing.
12. Why is Text Mining Important?
13. Name some applications of web mining.
14. Name some Common Methods for Analyzing Text Mining.
15. Name the components of Multimedia mining architecture.

SECTION B

1. Explain the architecture of multimedia data mining.
2. Briefly discuss the advantages of web mining.
3. Briefly discuss the applications of web mining.
4. Explain the types of web mining.
5. Briefly discuss the Text Mining Techniques.
6. Explain the advantages and disadvantages of text mining.
7. Define text mining and explain why it is important.
8. Explain Categories of Multimedia Data Mining.
9. Differentiate Spatial and Temporal data mining.
10. Explain the types of Multimedia Applications.

SECTION C

1. Describe Spatial and Temporal data mining in detail.
2. Explain in detail about web mining.
3. Describe text mining with its techniques in detail
4. Explain in detail about Multimedia Data Mining.
5. Explain in detail about types and categories of multimedia data mining.

